EDUGENIE – PROJECT DEVELOPMENT PHASES

3. Project Design Phase

The project design phase defines the architecture and structure of the EduGenie system.

System Architecture

EduGenie follows a modular client-server architecture.

Main Components

User → HTML/CSS Frontend → FastAPI Backend → AI Modules → Gemini/Local AI → Response → User

The frontend collects the user's request and sends it to the FastAPI backend. The backend identifies the requested operation and forwards it to the appropriate module.

Major Modules

1. **Q&A Module**
2. **Explanation Module**
3. **Quiz Module**
4. **Summary Module**
5. **Learning Path Module**

Folder Structure

```
EduGenie/
│
├── main.py
├── explanation_module.py
├── qna.py
├── quiz_module.py
├── summary_module.py
├── learning_path.py
├── requirements.txt
│
├── templates/
│   └── index.html
│
└── static/
    └── style.css
```

API Design

The following REST API endpoints were planned:

Endpoint	Purpose
`/qa`	Answer questions
`/explain`	Explain concepts
`/quiz`	Generate quizzes
`/summarize`	Summarize text
`/learn/recommendati`	Generate learning paths

Endpoint	Purpose
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This modular design makes the system easier to maintain and extend.